SIMATS ENGINEERING
CO2_ASSESSMENT TOOL3_Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct
I M.Kyathi Sai Priya (192572101) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: ____M.Kyathi__

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge

servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

Ans : Cloud Auto-Scaling and Load Balancing for Seasonal Sales

Interpretation of the Problem

During seasonal sales (such as Diwali, Black Friday, or festive offers), an e-commerce company receives a sudden increase in website traffic. If the cloud infrastructure cannot handle the extra users, the website may become slow or unavailable. Cloud auto-scaling, load balancing, and monitoring tools help maintain smooth performance.

Explanation

Cloud Auto-Scaling

Auto-scaling is a cloud feature that automatically increases or decreases the number of servers based on the website traffic. When many users visit the website, additional servers are created to handle the extra load. Once the traffic reduces, the extra servers are removed automatically. This ensures that the website always has enough resources without wasting money on unused servers.

Load Balancing

Load balancing works together with auto-scaling by distributing incoming user requests evenly among all available servers. Instead of sending all requests to one server, it shares the workload equally. This prevents any single server from becoming overloaded. If one server stops working, the load balancer automatically redirects users to the remaining healthy servers, ensuring continuous service.

Monitoring Tools

Monitoring tools continuously observe the performance of the cloud infrastructure. They track important parameters such as CPU usage, memory usage, network traffic, response time, and server health. When these values exceed a predefined limit, the monitoring system automatically triggers auto-scaling to add more servers. It also sends alerts to administrators if any problem occurs, helping them identify and fix issues quickly.

How This Reduces Downtime

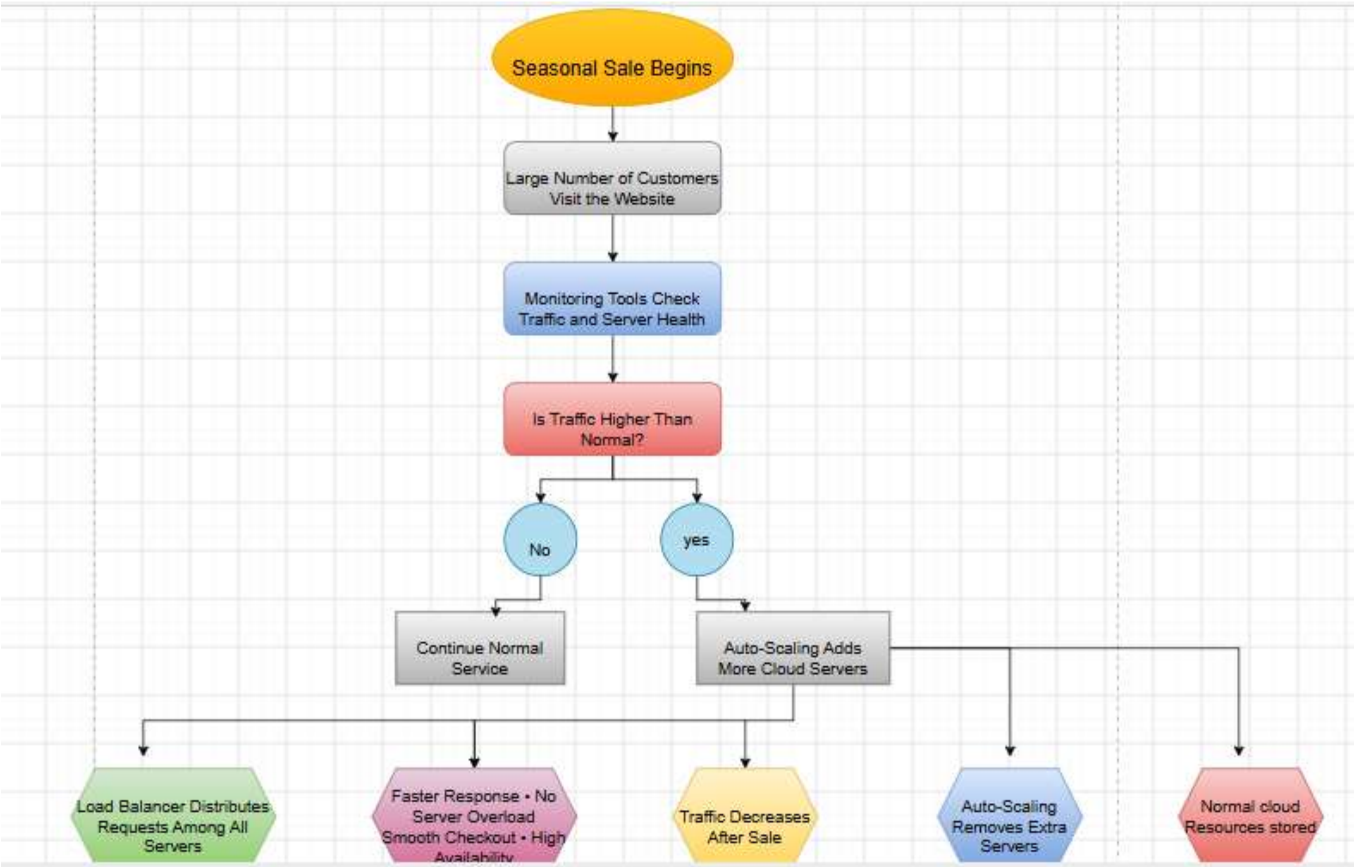
The combination of auto-scaling and load balancing keeps the website available even during heavy traffic. New servers are added automatically before the existing servers become overloaded. Load balancing distributes requests efficiently, and if a server fails, traffic is redirected to healthy servers. As a result, users experience very little or no downtime.

How This Improves User Experience

Customers can access the website without delays, pages load faster, products can be searched easily, and payments are completed smoothly. Since the website remains stable even during peak sales, customers enjoy a better shopping experience, leading to higher satisfaction and increased sales for the company.

Real-World E-Commerce Scenario

A good example is the **Amazon Great Indian Festival** or **Flipkart Big Billion Days** sale. During these events, millions of customers visit the website at the same time. The cloud monitoring system detects the increase in traffic and automatically launches additional servers. The load balancer distributes customer requests among all available servers so that no single server is overloaded. After the sale ends and the number of users decreases, the extra servers are removed automatically. This allows customers to browse products, place orders, and complete payments without interruptions while helping the company reduce unnecessary cloud costs.



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Ans : Cloud Storage for Secure Backup Data

Introduction

Many organizations generate a large amount of backup data every day. Storing this data on local servers is expensive and risky because hardware failures, natural disasters, or cyberattacks can cause data loss. Cloud storage provides a secure, reliable, and cost-effective solution by storing backup data across multiple data centers.

How Cloud Storage Ensures Durability and Data Redundancy

1. Durability

Durability means the backup data remains safe and available for many years without being lost.

Cloud providers achieve durability by:

- Storing multiple copies of data.
- Performing automatic error detection and correction.
- Replacing damaged storage devices automatically.
- Regularly checking data integrity.

Result: Even if one storage disk fails, the data is still available from another copy.

2. Data Redundancy

Data redundancy means keeping duplicate copies of the same data in different servers or locations.

Cloud storage provides:

- **Replication:** Data is copied to multiple servers.
- **Multi-zone storage:** Copies are stored in different availability zones.
- **Multi-region storage:** Copies are stored in different geographic regions.

If one server or an entire data center fails, another copy is used immediately.

How Encryption and Access Control Protect Backup Data

1. Encryption

Encryption converts readable data into unreadable data.

Two types are commonly used:

a) Data at Rest

- Data is encrypted while stored in cloud storage.
- Even if someone accesses the storage, they cannot read the files without the encryption key.

b) Data in Transit

- Data is encrypted while being transferred between the user and the cloud using secure protocols such as HTTPS/TLS.
- Prevents hackers from intercepting backup files.

2. Access Control

Access control ensures only authorized users can access backup data.

Common methods include:

- Username and password
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)
- Permission policies
- Identity and Access Management (IAM)

Benefits

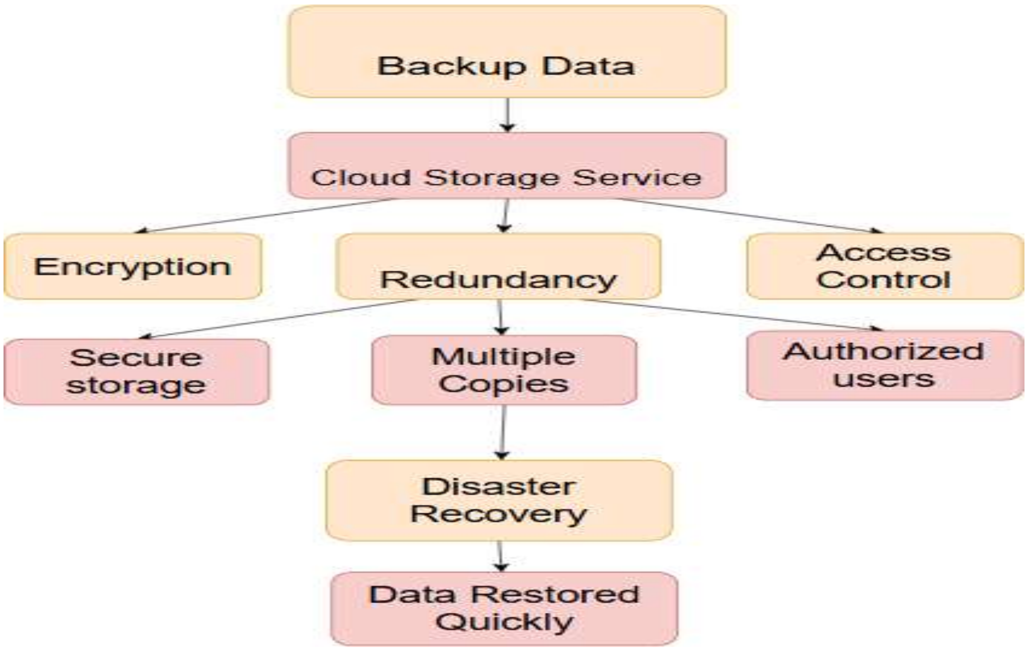
- Prevents unauthorized access.
- Protects confidential information.
- Tracks user activities through logs.

Cost Advantages

- Pay-as-you-go pricing model.
- No need to purchase expensive storage hardware.
- Lower maintenance and electricity costs.
- Easy scalability without additional infrastructure investment.

Disaster Recovery Example

A company stores daily database backups in the cloud. If its local data center is damaged by a flood or fire, it restores all critical data from the cloud and resumes business operations quickly.



3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

Ans :

1. How Content Delivery Networks (CDNs) Improve Performance and Reduce Latency

A **Content Delivery Network (CDN)** is a network of servers distributed across different countries and regions. Instead of sending data from one central server, the CDN stores copies of content on multiple servers worldwide.

Working

- User requests app content (video, image, file, webpage).
- The request is directed to the **nearest CDN server**.
- The nearby server sends the content instead of the distant origin server.

Benefits

- Reduces loading time.
- Minimizes network latency.
- Decreases traffic on the main server.
- Improves application speed during peak usage.
- Increases reliability by serving content even if one server is busy.

2. How Edge Servers Enhance User Experience

Edge servers are servers located close to end users.

Advantages

- Faster response because data travels a shorter distance.
- Smooth video playback with less buffering.
- Lower delay in multiplayer games.
- Faster app updates and downloads.
- Better performance during heavy traffic.

Example

- A user in **India** accesses the app.
- Instead of connecting to a server in the **USA**, the request is served from an edge server in **Mumbai** or **Chennai**, reducing response time.

3. How Cloud Providers Support Global Availability

Major cloud providers deploy applications across multiple regions and availability zones.

They provide:

- **Multiple data centers** across the world.
- **Automatic load balancing** to distribute traffic.
- **Auto-scaling** to handle increasing users.
- **Data replication** so copies exist in several regions.
- **Automatic failover** if one region becomes unavailable.

Benefits

- High availability (24×7 service).
- Better fault tolerance.
- Faster access for global users.
- Reduced downtime.

4. Real-Time Streaming / Gaming Example

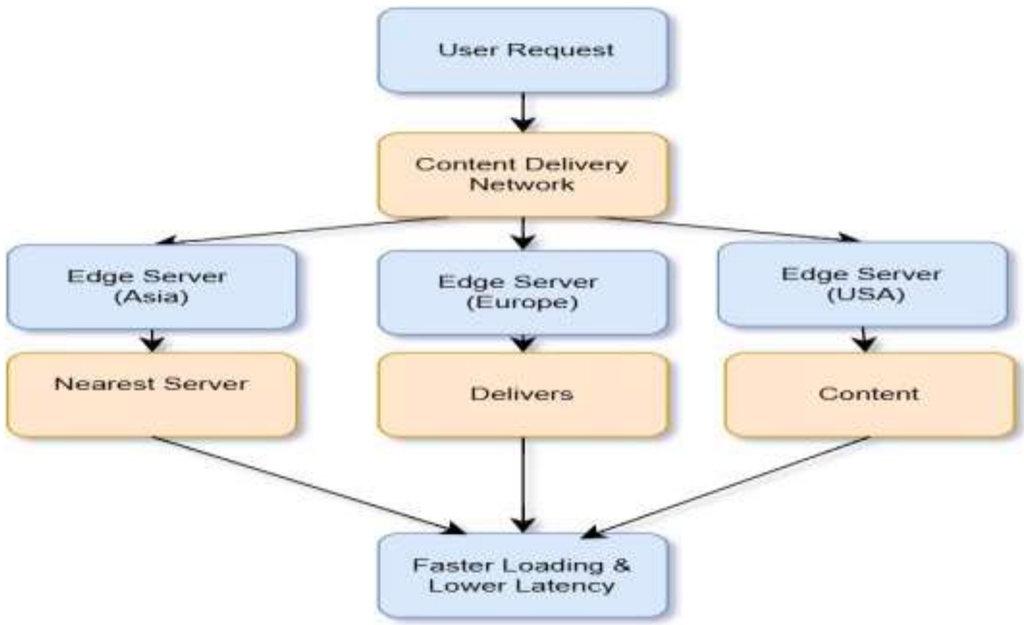
Example: Live Video Streaming Platform

How it works

- The live stream is uploaded once to the origin server.
- CDN distributes the stream to edge servers worldwide.
- Users watch the stream from the nearest edge server.

- Result:
    - Minimal buffering.
    - Low latency.
    - Smooth HD streaming.
    - Supports millions of viewers simultaneously.

The same approach benefits **online gaming**, where game assets and updates are delivered from nearby edge servers, reducing lag and improving gameplay.



Advantages

- Faster content delivery.
- Reduced latency.
- Improved user experience.
- Lower load on the origin server.
- High availability and reliability.
- Better scalability for global users.
- Reduced buffering and gaming lag.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

Ans : Cloud Security and Compliance for a Financial Institution

Financial institutions handle highly sensitive customer and transaction data. To protect this information and comply with government regulations, cloud providers offer **security, compliance, auditing, encryption, and monitoring** features.

1. How Cloud Providers Support Regulatory Compliance and Auditing

Cloud providers help organizations meet industry regulations by providing certified and secure cloud services.

They support:

- Compliance with standards such as **PCI DSS**, **ISO 27001**, **SOC 2**, and local financial regulations.
- Regular security audits and compliance reports.
- Identity and Access Management (IAM) to control who can access data.
- Audit trails that record every user activity.
- Automated compliance checks and policy enforcement.

Benefits:

- Ensures legal and regulatory compliance.
- Simplifies audits.
- Protects sensitive financial information.

2. Role of Logging, Monitoring, and Encryption

a) Logging

Logging records every activity performed in the cloud.

Examples:

- User login/logout
- Data access
- File modifications
- Transaction history

Benefits:

- Detects unauthorized access.
- Helps investigate security incidents.
- Provides audit evidence.

b) Monitoring

Monitoring continuously observes cloud resources and user activities.

Examples:

- Detect unusual login attempts.
- Monitor server performance.
- Alert administrators about suspicious behavior.

Benefits:

- Early detection of cyberattacks.
- Reduces downtime.
- Improves overall security.

c) Encryption

Encryption converts readable data into an unreadable format.

Two types:

- **Data at Rest:** Encrypts stored data.
- **Data in Transit:** Encrypts data while it travels over the internet.

Benefits:

- Prevents unauthorized access.
- Protects confidential customer and banking information.
- Ensures secure communication.

3. Ensuring Data Sovereignty Requirements

Data sovereignty means data must remain within a specific country or region according to local laws.

The company can ensure this by:

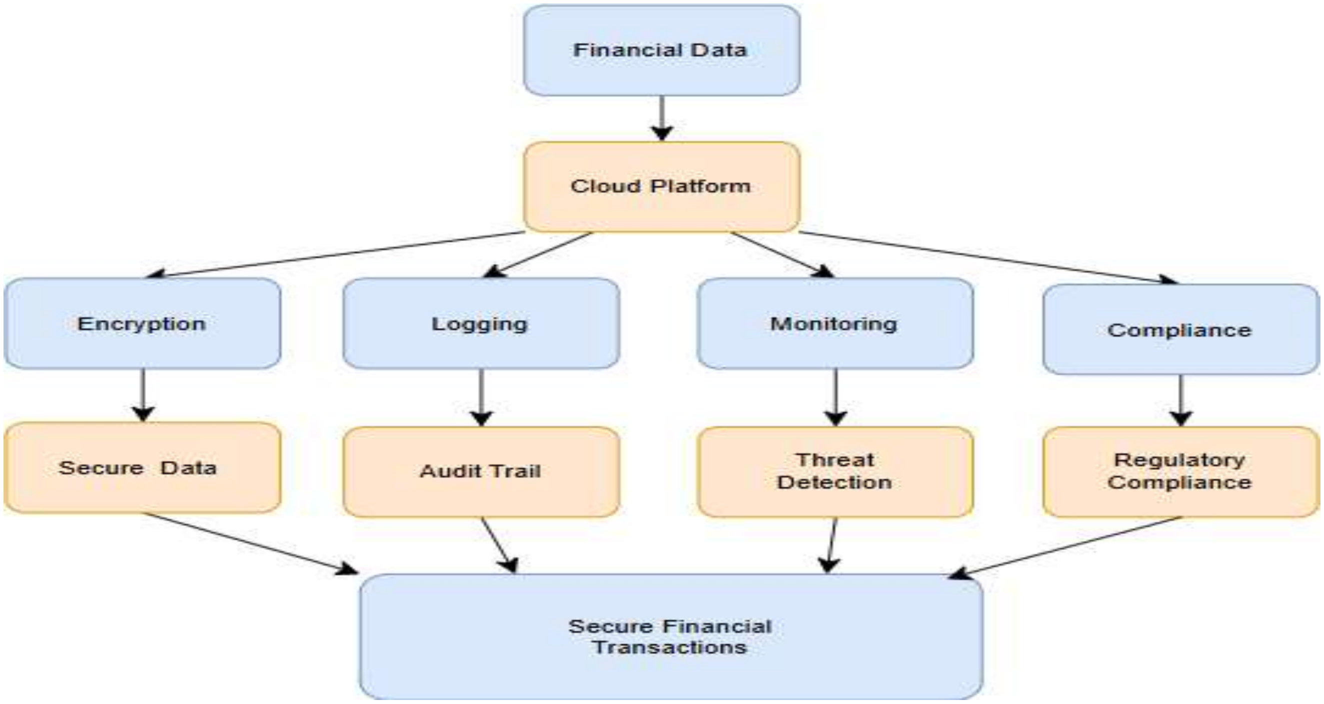
- Choosing cloud regions located within the required country.
- Storing customer databases only in approved data centers.
- Restricting cross-border data transfers.
- Applying regional backup and disaster recovery policies.
- Regularly auditing data location and access.

Benefits:

- Meets government regulations.
- Protects customer privacy.
- Avoids legal penalties.

4. Example: Sensitive Financial Transactions

A bank processes online fund transfers using a cloud platform.



How Security Is Maintained

- Customer logs in using secure authentication.
- Transaction data is encrypted before transmission.
- Every transaction is recorded in logs.
- Monitoring tools detect unusual activities, such as multiple failed login attempts.
- Customer data is stored only in the required geographic region to satisfy data sovereignty rules.
- Audit reports help regulators verify compliance.

5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Ans : Containerization for Application Deployment

A development team can use **containers** to package an application along with its dependencies, making it easy to deploy and run consistently on any system. Containers are lightweight, fast, and ideal for modern cloud applications.

Difference Between Containers and Virtual Machines (Theory)

Containers

Containers are a lightweight virtualization technology that packages an application along with its libraries, dependencies, and configuration files. They **share the host operating system's kernel**, so they do not require a separate operating system for each application. As a result, containers start quickly, consume fewer CPU, memory, and storage resources, and provide high performance. They are ideal for cloud-native applications, microservices, and DevOps environments because they are portable and can run consistently across different platforms.

Virtual Machines (VMs)

Virtual Machines are hardware-level virtualization environments that emulate a complete computer system. Each virtual machine includes its **own guest operating system**, application, and required libraries. Since every VM runs a separate operating system, they require more CPU, memory, and storage resources than containers. VMs take longer to boot but provide stronger isolation and are suitable for running different operating systems on the same physical machine or for applications requiring high security and isolation.

Theoretical Differences

1. Architecture

- **Containers:** Share the host operating system kernel and run isolated applications.
- **Virtual Machines:** Run on a hypervisor, with each VM having its own complete operating system.

2. Performance

- **Containers:** Offer faster performance because they share the host OS and have less overhead.
- **Virtual Machines:** Performance is slightly slower due to the overhead of running multiple operating systems.

3. Resource Usage

- **Containers:** Consume less CPU, memory, and storage, allowing many containers to run on a single server.
- **Virtual Machines:** Require more resources because each VM needs its own operating system.

4. Startup Time

- **Containers:** Start within a few seconds since only the application and its dependencies are loaded.
- **Virtual Machines:** Take several minutes to boot because the entire operating system must start.

5. Portability

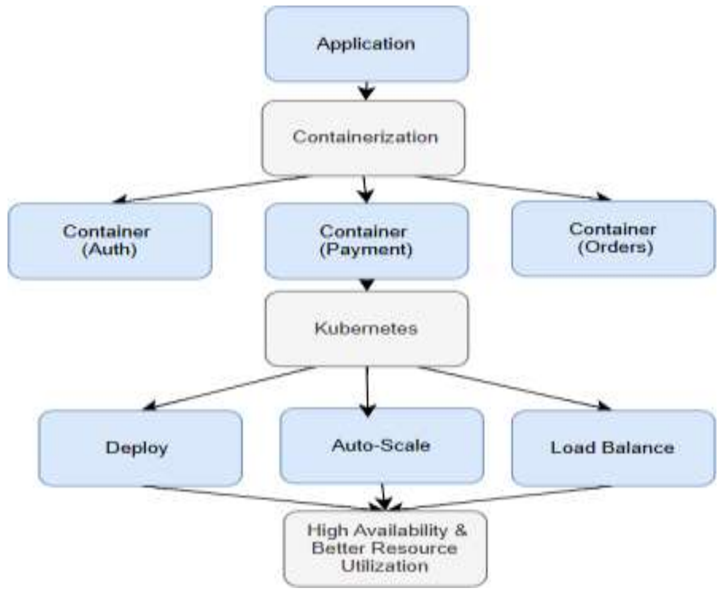
- **Containers:** Highly portable and can run consistently across different cloud platforms and operating systems that support containers.
- **Virtual Machines:** Less portable because VM images are larger and may require compatible hypervisors.

6. Security and Isolation

- **Containers:** Provide process-level isolation but share the host operating system, offering slightly less isolation.
- **Virtual Machines:** Provide stronger isolation because each VM has its own operating system and virtual hardware.

7. Typical Use Cases

- **Containers:** Best suited for microservices, cloud-native applications, CI/CD pipelines, and scalable web applications.
- **Virtual Machines:** Best suited for legacy applications, running multiple operating systems, and workloads requiring strong isolation.



Explanation

- **Containers** share the host operating system, making them lightweight and efficient.
- **Virtual Machines** run a complete operating system for each instance, requiring more memory and storage.

2. How Kubernetes Simplifies Deployment and Scaling

Kubernetes is a container orchestration platform that automates the management of containers.

Kubernetes Features

- Automatically deploys containers.
- Scales applications up or down based on demand.
- Balances traffic across containers.

- Restarts failed containers automatically.
- Performs rolling updates without downtime.
- Monitors application health.

Benefits

- Easy deployment.
- High availability.
- Automatic scaling.
- Reduced manual effort.
- Better resource utilization.

3. Portability Across Different Cloud Environments

Containers package the application and all required libraries together.

This allows the same container to run on:

- Public cloud (AWS, Azure, Google Cloud)
- Private cloud
- Hybrid cloud
- Local development machines

Benefits

- "Build once, run anywhere."
- Easy migration between cloud providers.
- Consistent application behavior.
- Reduced compatibility issues.
- Faster software delivery.

How It Works

- Each service runs inside its own container.
- Kubernetes deploys and manages all containers.
- If user traffic increases, Kubernetes automatically creates more container instances.
- If one container fails, Kubernetes restarts it automatically.

- The same application can run on AWS, Azure, Google Cloud, or a private cloud without code changes.

Advantages of Containers

- Faster startup.
- Lightweight and efficient.
- Lower resource usage.
- Easy deployment.
- Automatic scaling with Kubernetes.
- High portability.
- Better fault isolation.
- Supports continuous integration and continuous deployment (CI/CD).